

Hands-On Projects

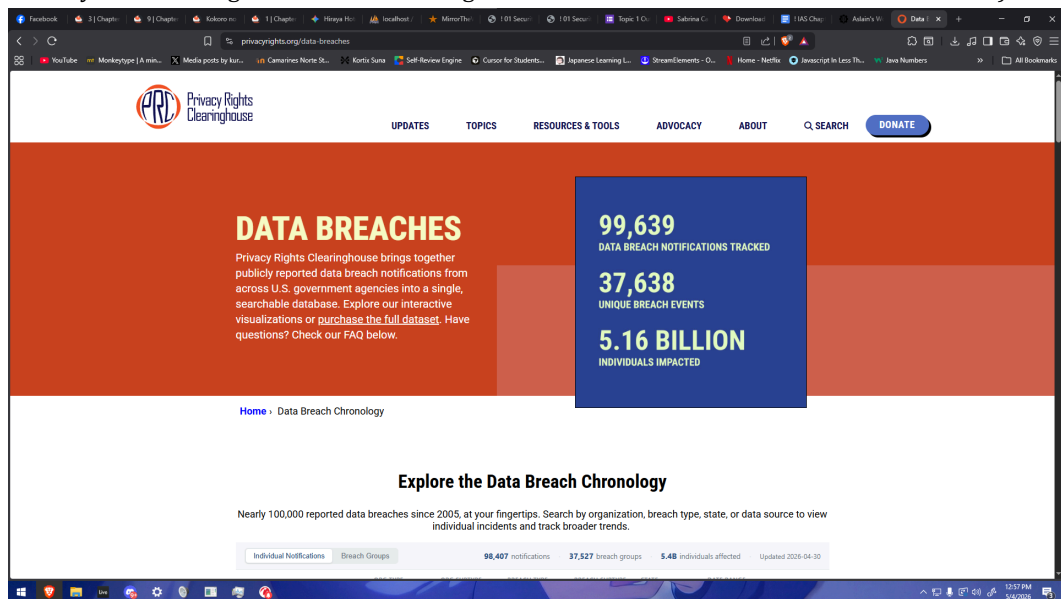


Project 1-1: Examine Data Breaches—Textual

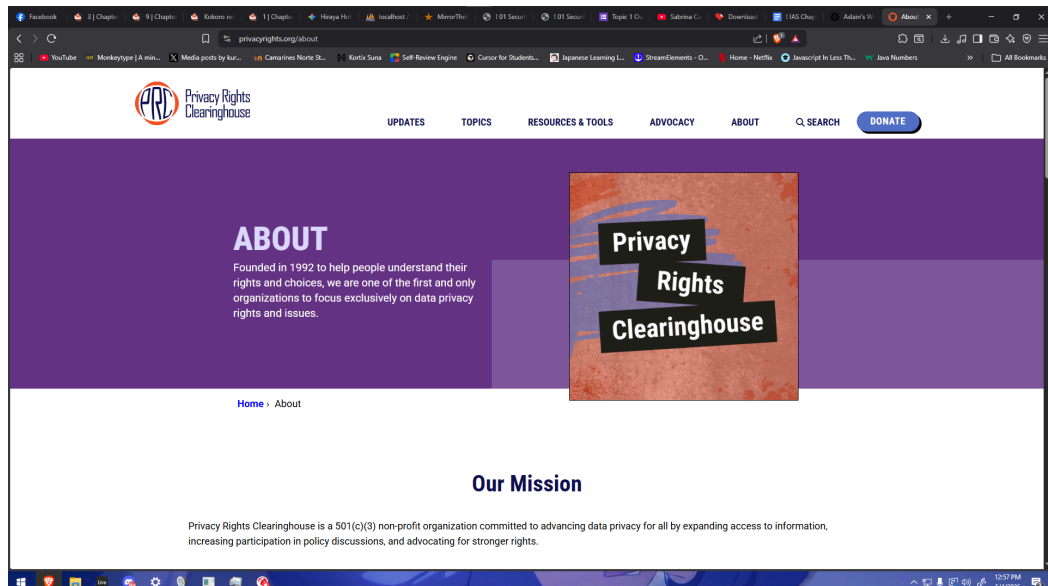
The Privacy Rights Clearinghouse (PRC) is a nonprofit organization whose goals are to raise consumers' awareness of how technology affects personal privacy and empower consumers to take action to control their personal information. The PRC maintains a searchable database of security breaches that

impact consumer's privacy. In this project you will gather information from the PRC website.

1. Open a web browser and enter the URL www.privacyrights.org/data-breach (if you are no longer able to access the site through the web address, use a search engine to search for "Privacy Rights Clearinghouse data breach").

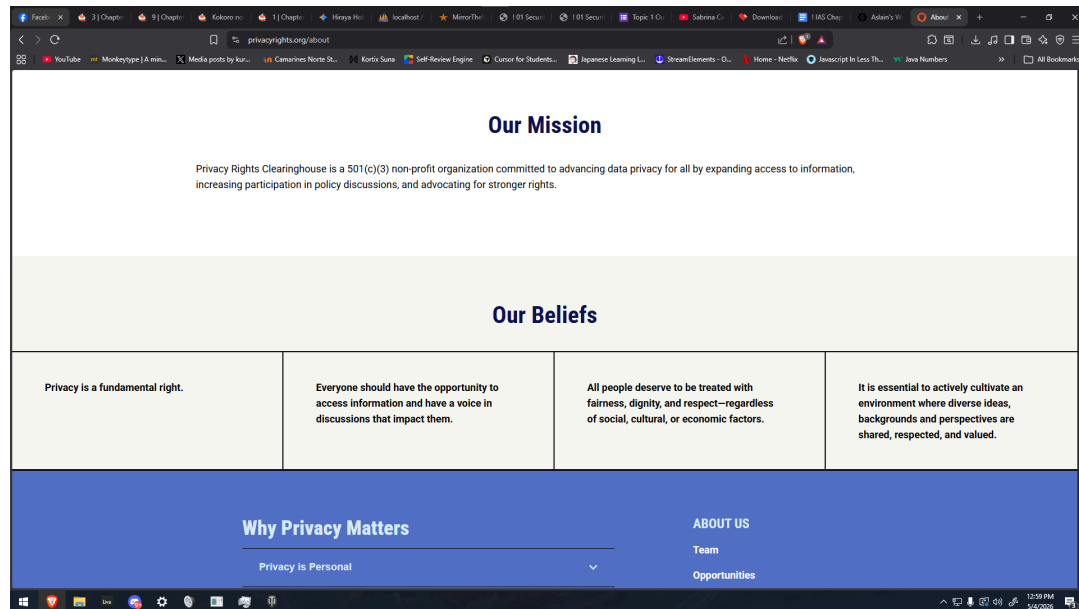


2. first spend time reading about the PRC. Click About Us in the toolbar.

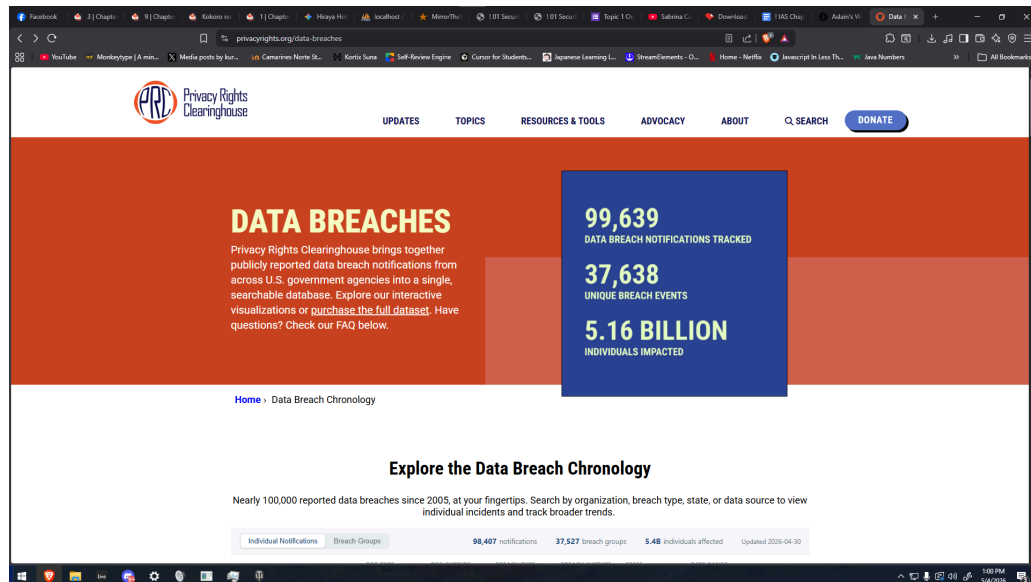


- 3.

3. Scroll down to the content under Mission and Goals and also under Services. Spend a few minutes reading about the PRC.



4. Click your browser's Back button to return to the previous page.



5. On the Chronology of Data Breaches page scroll down and observe the different breaches listed in chronological order.



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Organization	Reported Date	Breach Date	Org Type	Breach Type	Impact	Location	Source
Humana Inc.	Apr 30, 2026	Unknown	Medical/Health...	Hacking/IT Inc...	—	UNKN	ID
TransGlobal Insurance Agency	Apr 29, 2026	Feb 10, 2026	Business (Finan...	Hacking/IT Inc...	—	UNKN	CA
Credit Technologies, Inc.	Apr 28, 2026	Unknown	Business (Finan...	Unknown	—	UNKN	MA
Higginbotham Insurance Agency	Apr 28, 2026	Unknown	Business (Finan...	Unknown	—	UNKN	MA
Lincoln Holdings LLC d/b/a Mo	Apr 28, 2026	Dec 6, 2025	Business (Other)	Hacking/IT Inc...	—	Washingt...	NE
JC Resorts LLC	Apr 28, 2026	Jan 19, 2026	Business (Other)	Hacking/IT Inc...	—	La Jolla, CA	NE
Frontwave Credit Union	Apr 28, 2026	Apr 3, 2026	Business (Finan...	Unintended Dis...	—	UNKN	CA
Churchill Claims Services, Inc.	Apr 28, 2026	Unknown	Business (Finan...	Unknown	—	UNKN	MA
U.S. Bank, N.A.	Apr 28, 2026	Unknown	Business (Finan...	Unknown	—	UNKN	MA

- Now create a customized list of the data that will only list data breaches of educational institutions. Scroll back to the top of the page.
- Under Select organization type(s), uncheck all organizations except EDU - Educational Institutions.

Organization	Breach Type	Impact
The University of Massachusetts Amherst	UNKN	—
Texas Tech University Health Sciences Center	UNKN	729K
Eanes Independent School District	UNKN	10K
Texas Tech University Health Sciences Center	UNKN	814K
Texas Tech University Health Sciences Center	HACK	—
Abby Kelley Charter Foster Public School	UNKN	—
Washington International School	UNKN	—
Empowerment Schools - Healthcare Ltd and Texas Medical Careers, Limited	UNKN	69K
Maine Community College System	DISC	8
The College of Health Care Professions	HACK	—
Empowerment Schools - Healthcare Ltd and Texas Medical Careers, Limited	UNKN	—

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- Click GO!
- Scroll down to Breach Subtotal if necessary. How many breaches that were made public pertain to educational institutions?
37,527
- Scroll down and observe the breaches for educational institutions.
- Scroll back to the top of the page. Click New Search, located beneath the GO! button.
- Now search for breaches that were a result of lost, discarded, or stolen equipment that belonged to the government and military. Under Choose the type of breaches to display, uncheck all types except Portable device (PORT) - Lost, discarded or stolen lap-top, PDA, smartphone, portable memory device, CD, hard drive, data tape, etc.

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Explore the Data Breach Chronology

Nearly 100,000 reported data breaches since 2005, at your fingertips. Search by organization, breach type, state, or data source to view individual incidents and track broader trends.

Individual Notifications | Breach Groups | 98,407 notifications | 37,527 breach groups | 5.48 individuals affected | Updated 2025-04-30

SEARCH: Organization, vendor, incident detail | Org Type: Government (2) | Org Subtype: All Subtypes | Breach Type: Portable Device | Breach Subtype: All Subtypes | State: All States | Date Range: All Time | Search

Organization	Breach Type	Impact
Montana Fish, Wildlife and Parks	PORT	—
San Francisco VA Health Care System	PORT	941
Fire Department, City of New York	PORT	10K
City of Philadelphia	PORT	2K
City of Georgetown Fire Department	PORT	1K
Seattle Housing Authority	PORT	720
Washington State Patrol	PORT	2K
California Department of Public Health	PORT	—
Wisconsin Department of Health Services	PORT	779
City of Detroit	PORT	544
Mann-Grandstaff VA Medical Center	PORT	2K

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15. Under Select organization type(s), uncheck all organizations except GOV — Government and Military.

14. Click GO!.

5. Scroll down to Breach Subtotal, if necessary. How many breaches that were made public pertain to this type?

1456 breach total

16. Scroll down and observe the breaches for governmental institutions.

17. Scroll back to the top of the page.
18. Now create a search based on criteria that you are interested in, such as the Payment Card Fraud against Retail/Merchants during the current year.

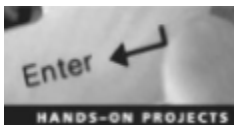
1

The screenshot shows the privacyrights.org/data-breaches website. It features a search bar with filters for Organization, vendor, incident date, Org Type, Org Subtype, Breach Type, Breach Subtype, State, and Date Range. Below the search bar is a table of data breaches.

Organization	Reported Date	Breach Date	Org Type	Breach Type	Impact	Location	Source
Brandt Equities LP	Apr 24, 2026	Unknown	Business (Retail)	Unknown	—	UNKN	MA
Rev Up Brands, LLC d/b/a Rev	Apr 21, 2026	Mar 16, 2025	Business (Retail)	Hacking/IT Inc...	—	Niles, IL	NH
Murray's Cheese	Apr 21, 2026	Unknown	Business (Retail)	Unknown	—	UNKN	MA
Rev Up Brands, LLC d/b/a Rev	Apr 18, 2026	Unknown	Business (Retail)	Unknown	—	UNKN	MA
Rev Up Brands, LLC d/b/a Rev	Apr 17, 2026	Mar 16, 2025	Business (Retail)	Hacking/IT Inc...	5,841	Niles, IL	ME
DollarDays International, Inc.	Apr 16, 2026	Unknown	Business (Retail)	Unknown	—	UNKN	MA
Mullinax Automotive d/b/a Ms	Apr 15, 2026	Dec 4, 2025	Business (Retail)	Hacking/IT Inc...	2	Apogee, FL	ME
Mullinax Ford of Central Florida	Apr 15, 2026	Unknown	Business (Retail)	Unknown	—	UNKN	MA
Hoshino (U.S.A.) Inc.	Apr 15, 2026	Unknown	Business (Retail)	Unknown	—	UNKN	MA

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- 19.
20. When finished, close all windows.



Project 1-2: Examine Data Breaches—Visual

In this project you will view the biggest data breaches resulting in stolen information through a visual format.

1. Open your web browser and enter the URL <http://www.informationisbeautiful.net/visualizations/worlds-biggest-data-breaches-hacks/> (if you are no longer able to access the site through this web address, use a search engine to search for “Information Is Beautiful World's Biggest Data Breaches”).



2. Click Hide Filter to display a visual graphic of the data breaches, as shown in Figure I -6.

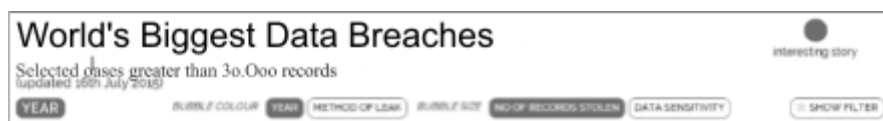
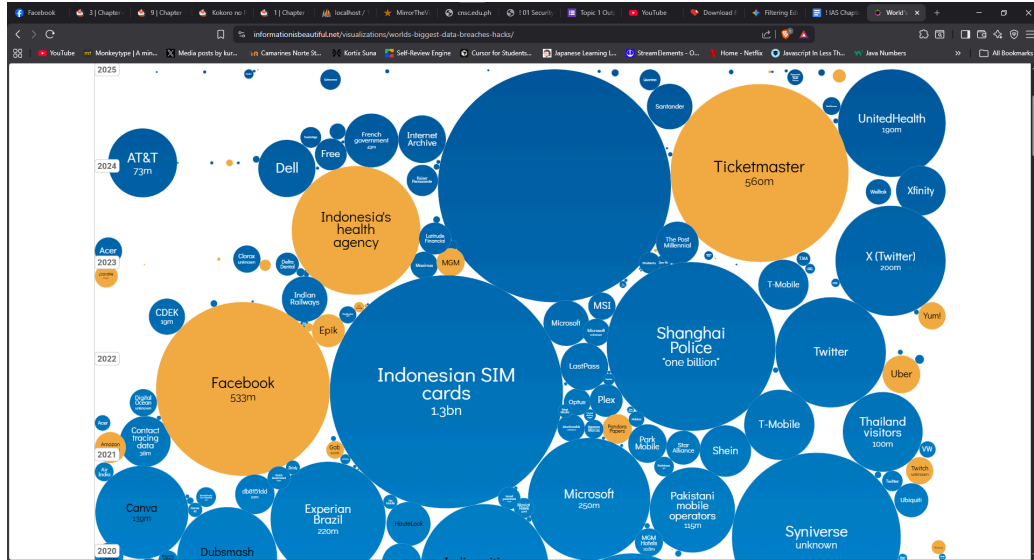


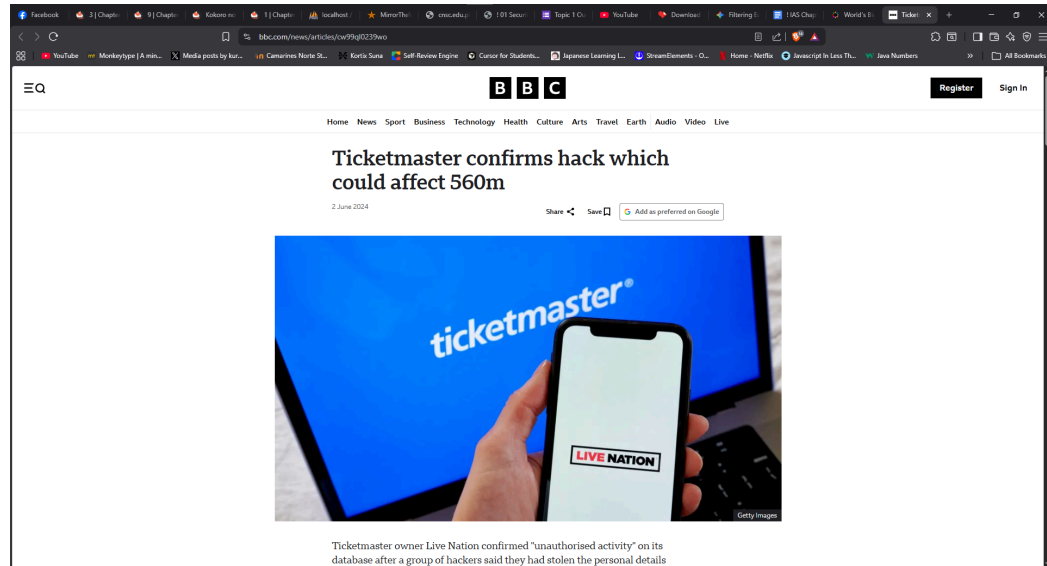
Figure 1-s wor|d's Biggest Data Breaches

Source: Information is Beautiful

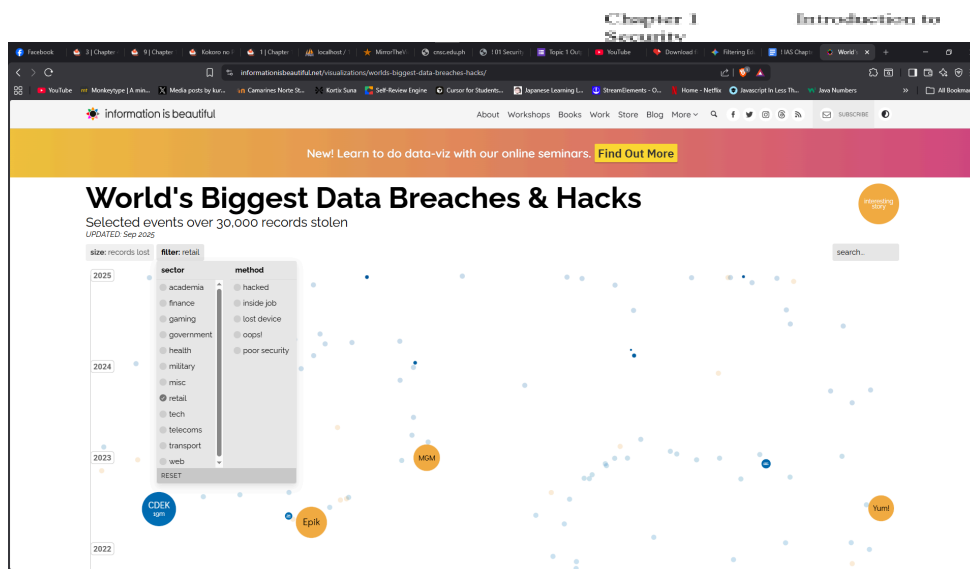
3. Scroll down the page to view the data breaches. Note that the size of the breach is indicated by the size of the bubble.



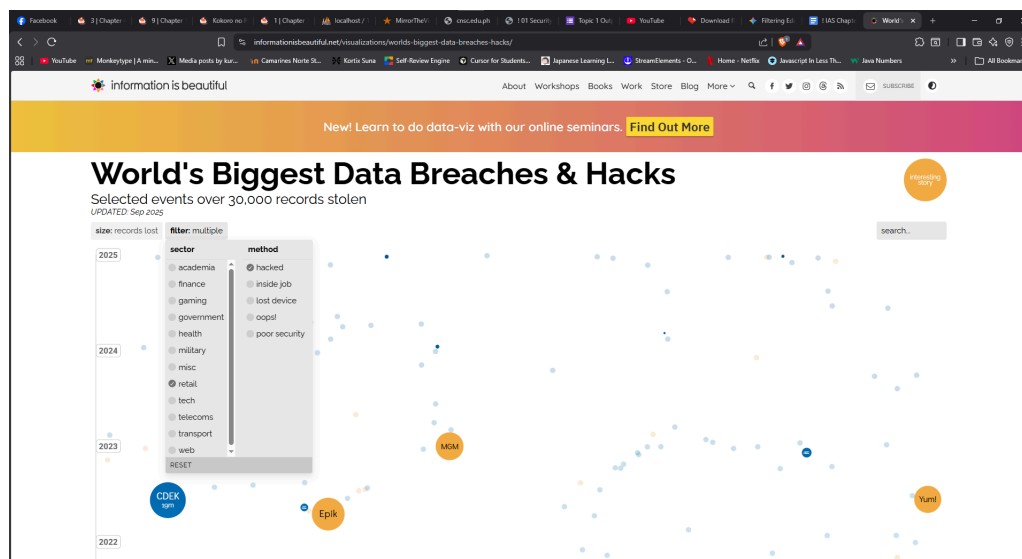
4. Scroll back up to the top and note that the color of the bubbles that have an “Interest-ing Story.” Click one of the bubbles and read the story.



5. Click Read a bit more.
6. Click Click to see the original report.
7. Read about the data breach. When finished close on ly this tab in your browser.
8. Click Show Filter to display the filter menu.
9. Under “Organisation” click Retail.



10. Under “Method of Leak” click Hacked. How many bubbles appear? 36



11. . Under “Organisation” click Government.



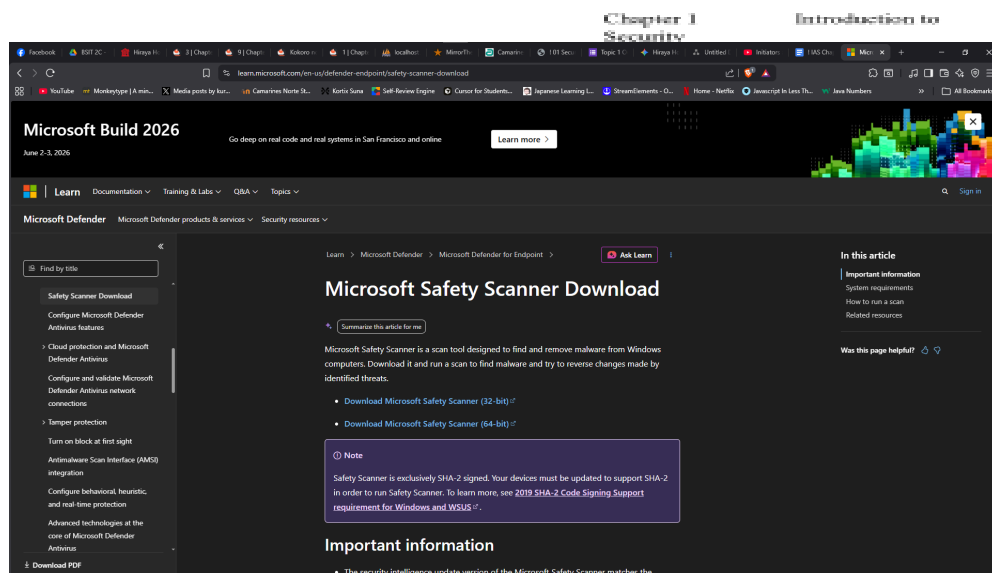
12. Under "Method of Leak" click Inside Job. How many bubbles appear? 2
43. Under "Organisation" click All.
14. Under "Method of Leak" click All.
15. At the top of the graphic click Method of Leak so that the bubbles display how the leak occurred. Which type of leak is the most common? Why do you think this is the case?
Hacked is the most common since it's mostly vulnerable to small enterprises.
16. Does this visual convey a better story than the textual data in the previous project? **Yes**
17. Close all windows.



Project 1-3: Scan for Malware Using the Microsoft Safety Scanner

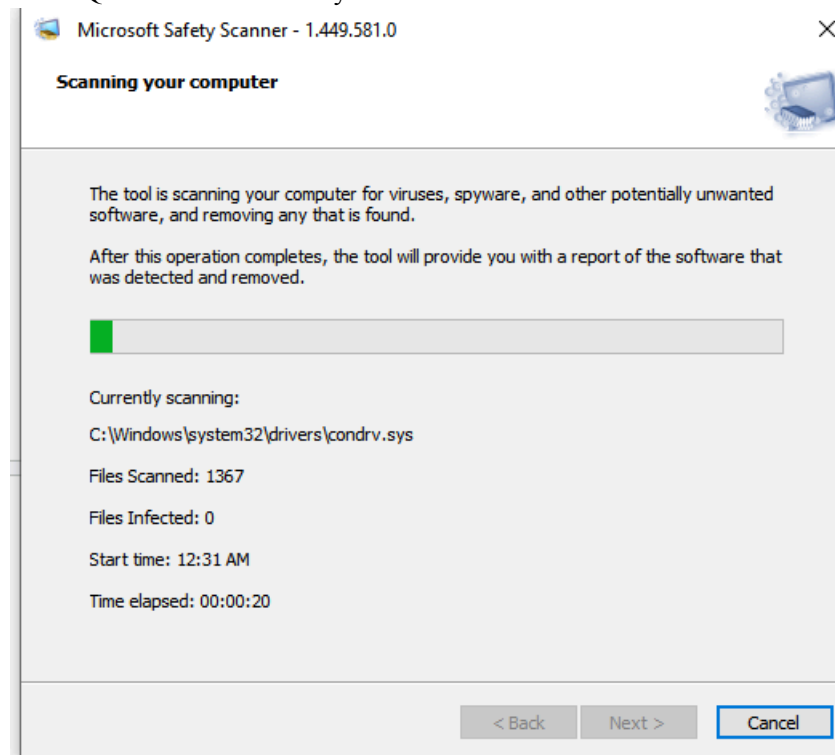
In this project you will download and run the Microsoft Safety Scanner to determine if there is any malware on the computer.

1. Determine which system type of Windows you are running. Click Start, Control Panel, System and Security, and then System. Look under System type for the description.
2. Open your web browser and enter the URL www.microsoft.com/security/scanner/en-us/default.asp (if you are no longer able to access the site through the URL, use a search engine to search for "Microsoft Safety Scanner").

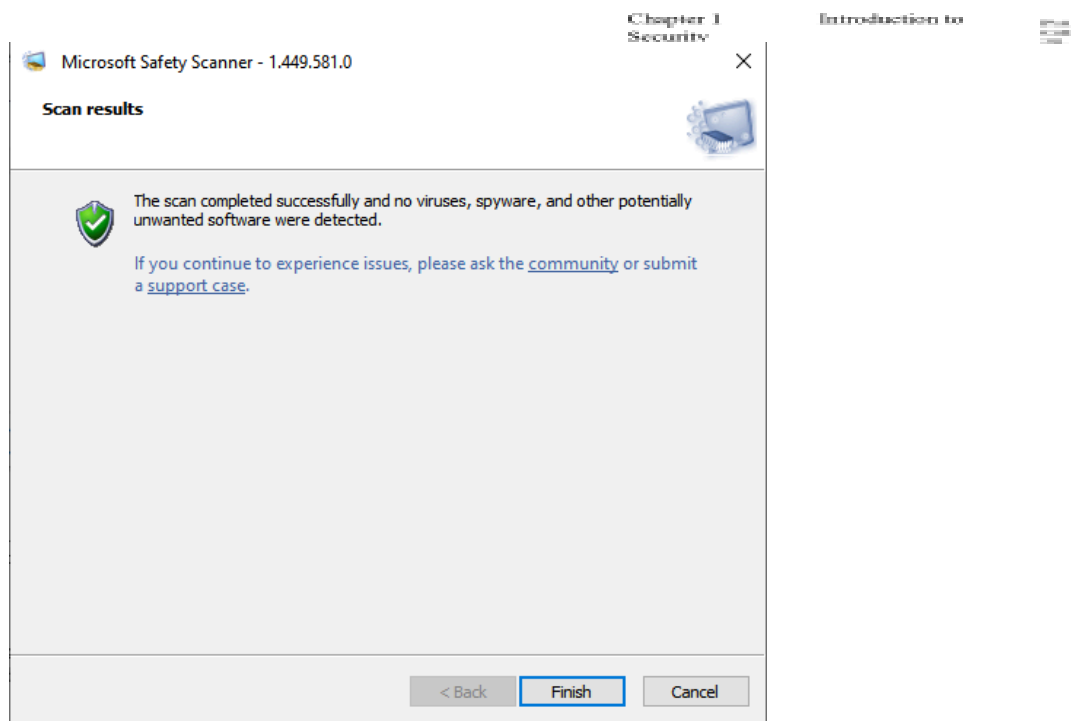


3. Click Download Now.
4. Select either 32-bit or 64-bit, depending upon which system type of Windows you are running.
5. When the program finishes downloading, right-click **Start** and click Open Windows Explorer.
6. Click the Downloads icon in the left pane.

7. Double-click the mserr.exe file.
8. Click Run. If the User Account Control dialog box appears, click Yes.
9. Click the check box to accept the license terms for this software. Click Next.
10. Click Next.
11. Select Quick scan if necessary.



12. Click Next.
13. Depending on your computer this scan may take several minutes. Analyze the results of the scan to determine if there is any malicious software found in your computer.
14. If you have problems you can click View detailed results of the scan. After reviewing the results, click OK. If you do not find any problems, click Finish.
15. If any malicious software was found on your computer run the scan again and select Full scan. After the scan is complete, click Finish to close the dialog box.



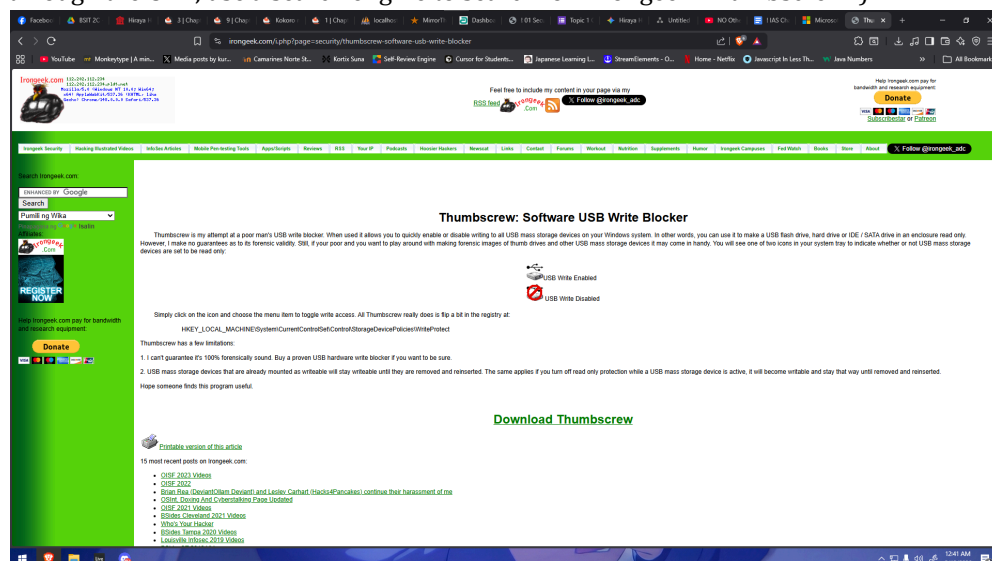
16. Close all windows.



Project 1-4: Write-Protecting a USB Flash Drive

Malicious software is often spread from one computer to another by infected USB flash drives. This can be controlled by write-protecting the drive so that no malware can be copied to it. In this project, you will download and install a software-based USB write blocker to prevent data from being written to a USB device. You will need a USB flash drive for this project.

1. Open your web browser and enter the URL www.irongeek.com/i.php?page=security/thumbscrew-software-usb-write-blocker (if you are no longer able to access the program through the URL, use a search engine to search for "Irongeek Thumbscrew").



2. Click Download Thumbscrew.
3. If the File Download dialog box appears, click Save and follow the instructions to save this file in a location such as your desktop or a folder designated by your instructor.



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4. When the file finishes downloading, extract the files in a location such as your desktop or a folder designated by your instructor. Navigate to that location and double-click thumbscrew.exe and follow the default installation procedures.
5. After installation, notice that a new icon appears in the system tray in the lower right corner of the screen.
6. Insert a USB flash drive into the computer.
7. Navigate to a document on the computer.
8. Right-click the document and then select Send to.
9. Click the appropriate Removable Disk icon of the USB flash drive to copy the file to the flash drive.